ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Angles in Polygons & Parallel Lines

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

ANGLES IN POLYGONS & PARALLEL LINES · CH4 · L2

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The Map

This lesson collapses Angles in Polygons & Parallel Lines into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Name the angle relationship

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

02 Use the polygon sum

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

03 Use exterior angles

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

04 Chain reasons in order

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

05 Find the gradient

TRIGGER *"work out", "find", a missing value*

06 Use trigonometry

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BANK EVIDENCE 20 website questions, 20 checked solutions, 4 local PDFs read.

01 Name the angle relationship

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Angles in Polygons & Parallel Lines question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$30 + (4x + 10) + (x + 20) = 180$$

SIMPLEST STRATEGY

1. Underline the target: what is the question asking you to find?
2. Write the rule, theorem, or equation before any calculator work.
3. Substitute one value at a time and keep units/unknowns visible.
4. Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2H Q4

Angles in a triangle add

$$30 + (4x + 10) + (x + 20) = 180$$

Finish: $x = 24$.

FORMULA BANK

Parallel lines: corresponding/alternate/co-interior. Polygon sum: $180(n-2)$. Exterior sum: $360°$.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 01

SAME IDEA Use the same first line from Strategy 01.

QUESTION

> 4 The diagram shows a triangle. Diagram NOT accurately drawn 30 degrees (4x + 10) degrees (x + 20) degrees Work out the value of x. x =

YOUR SOLUTION

02 Use the polygon sum

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Angles in Polygons & Parallel Lines question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$30 + (4x + 10) + (x + 20) = 180$$

SIMPLEST STRATEGY

1. Underline the target: what is the question asking you to find?
2. Write the rule, theorem, or equation before any calculator work.
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WORKED MODEL

Source pattern: Jan 2020 P2H Q4

Angles in a triangle add

$$30 + (4x + 10) + (x + 20) = 180$$

Finish: $x = 24$.

FORMULA BANK

Parallel lines: corresponding/alternate/co-interior. Polygon sum: $180(n - 2)$. Exterior sum: $360°$.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 02

SAME IDEA Use the same first line from Strategy 02.

QUESTION

> 4 The diagram shows a triangle. Diagram NOT accurately drawn 30 degrees (4x + 10) degrees (x + 20) degrees Work out the value of x. x =

YOUR SOLUTION

03 Use exterior angles

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Angles in Polygons & Parallel Lines question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$(9x + 27) + (2x - 23) = 180$$

SIMPLEST STRATEGY

1 Underline the target: what is the question asking you to find?
2 Write the rule, theorem, or equation before any calculator work.
3 Substitute one value at a time and keep units/unknowns visible.
4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Nov 2025 P1H Q12

The interior angle and exterior

$$(9x + 27) + (2x - 23) = 180$$

The exterior angle is

$$2x - 23 = 2(16) - 23 = 9^\circ$$

Finish: $n = 40$.

FORMULA BANK

Parallel lines: corresponding/alternate/co-interior. Polygon sum: $180(n-2)$. Exterior sum: 360°.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 03

SAME IDEA Use the same first line from Strategy 03.

QUESTION

> 12 Diagram NOT accurately drawn $(9x + 27)$ $(2x - 23)$ The diagram shows part of a regular n-sided polygon with an interior angle of $(9x + 27)$ degrees an exterior angle of $(2x - 23)$ degrees Work out the value of n n =

YOUR SOLUTION

04 Chain reasons in order

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Angles in Polygons & Parallel Lines question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$177,\ 175,\ 173,\ 171,\ 169, \ldots$$

SIMPLEST STRATEGY

1. Underline the target: what is the question asking you to find?
2. Write the rule, theorem, or equation before any calculator work.
3. Substitute one value at a time and keep units/unknowns visible.
4. Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jun 2022 P1H Q23

The interior angles form an

$$177,\ 175,\ 173,\ 171,\ 169, \ldots$$

Calculate value

$$a = 177, \quad d = -2$$

Finish: $n = 18$

FORMULA BANK

Parallel lines: corresponding/alternate/co-interior. Polygon sum: $180(n-2)$. Exterior sum: $360°$.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 04

SAME IDEA Use the same first line from Strategy 04.

QUESTION

> 23 A polygon has n sides, where $n > 5$ When arranged in order of size, starting with the largest number, the sizes of the interior angles of the polygon, in degrees, are the terms of an arithmetic sequence. Here are the first five terms of this sequence. 177 175 173 171 169 Find the value of n Show clear algebraic working. Question 23 continued n =

YOUR SOLUTION

05 Find the gradient

TRIGGER *"work out", "find", a missing value*

BECOMES A Angles in Polygons & Parallel Lines question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\frac{(5-2)180}{5} = 108^\circ$$

SIMPLEST STRATEGY

1. Underline the target: what is the question asking you to find?
2. Write the rule, theorem, or equation before any calculator work.
3. Substitute one value at a time and keep units/unknowns visible.
4. Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2021 P1HR Q8

A regular pentagon has interior

$$\frac{(5-2)180}{5} = 108^\circ$$

A regular hexagon has interior

$$\frac{(6-2)180}{6} = 120^\circ$$

Finish: 96°.

FORMULA BANK

Parallel lines: corresponding/alternate/co-interior. Polygon sum: $180(n-2)$. Exterior sum: 360°.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 05

SAME IDEA Use the same first line from Strategy 05.

QUESTION

> 8 The diagram shows a regular pentagon, ABCDE, a regular hexagon, CFGHID, and a quadrilateral, EDIJ. A B F G H C D I x J E Diagram NOT accurately drawn AEJ and HIJ are straight lines. Work out the size of the angle marked x. Show your working clearly. degrees

YOUR SOLUTION

06 Use trigonometry

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Angles in Polygons & Parallel Lines question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$177,\ 175,\ 173,\ 171,\ 169, \ldots$$

SIMPLEST STRATEGY

1. Underline the target: what is the question asking you to find?
2. Write the rule, theorem, or equation before any calculator work.
3. Substitute one value at a time and keep units/unknowns visible.
4. Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jun 2022 P1H Q23

The interior angles form an

$$177,\ 175,\ 173,\ 171,\ 169, \ldots$$

Calculate value

$$a = 177, \quad d = -2$$

Finish: $n = 18$

FORMULA BANK

Parallel lines: corresponding/alternate/co-interior. Polygon sum: $180(n-2)$. Exterior sum: $360°$.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 06

SAME IDEA Use the same first line from Strategy 06.

QUESTION

> 23 A polygon has n sides, where $n > 5$ When arranged in order of size, starting with the largest number, the sizes of the interior angles of the polygon, in degrees, are the terms of an arithmetic sequence. Here are the first five terms of this sequence. 177 175 173 171 169 Find the value of n Show clear algebraic working. Question 23 continued n =

YOUR SOLUTION

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"work out", "show", "hence", a familiar exam phrase	Name the angle relationship
"work out", "show", "hence", a familiar exam phrase	Use the polygon sum
"work out", "show", "hence", a familiar exam phrase	Use exterior angles
"work out", "show", "hence", a familiar exam phrase	Chain reasons in order
"work out", "find", a missing value	Find the gradient
"work out", "show", "hence", a familiar exam phrase	Use trigonometry

FINAL BOTTOM LINE Do not try to remember every past-paper wording. Remember the trigger, write the first line, and let the working follow.